

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — FULL STACK ACTIVE (BENCHMARKS 1-4)

Evaluating Agentic Architectures & Benchmarks

1. Benchmark Suite

- VerilogEval / Architecture
- OpenRTLSet Benchmarks
- ChiPBench Physical Runs
- VeriContaminated Suite

2. Evaluation Metric

- Pass@k Functional Rate
- PPA Pareto Dominance
- Contamination Auditing
- Verification Cost (USD)

3. Comparative Runs

- AI System vs Heuristics
- Equalized Event Budgets
- Unbiased Statistical Check
- Hardware Ground Truth

4. Benchmark Card

- Rigorous Scorecard
- Reproducibility Verdict
- Claim Qualification
- Handoff Validation

Benchmark Contamination & Audit Feedback Loop